

Mount Kigali



University

Empowering Generations Through Quality Education

Security and Cryptography (BIT8322) INTRODUCTION

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What is the difference between Data and Information?

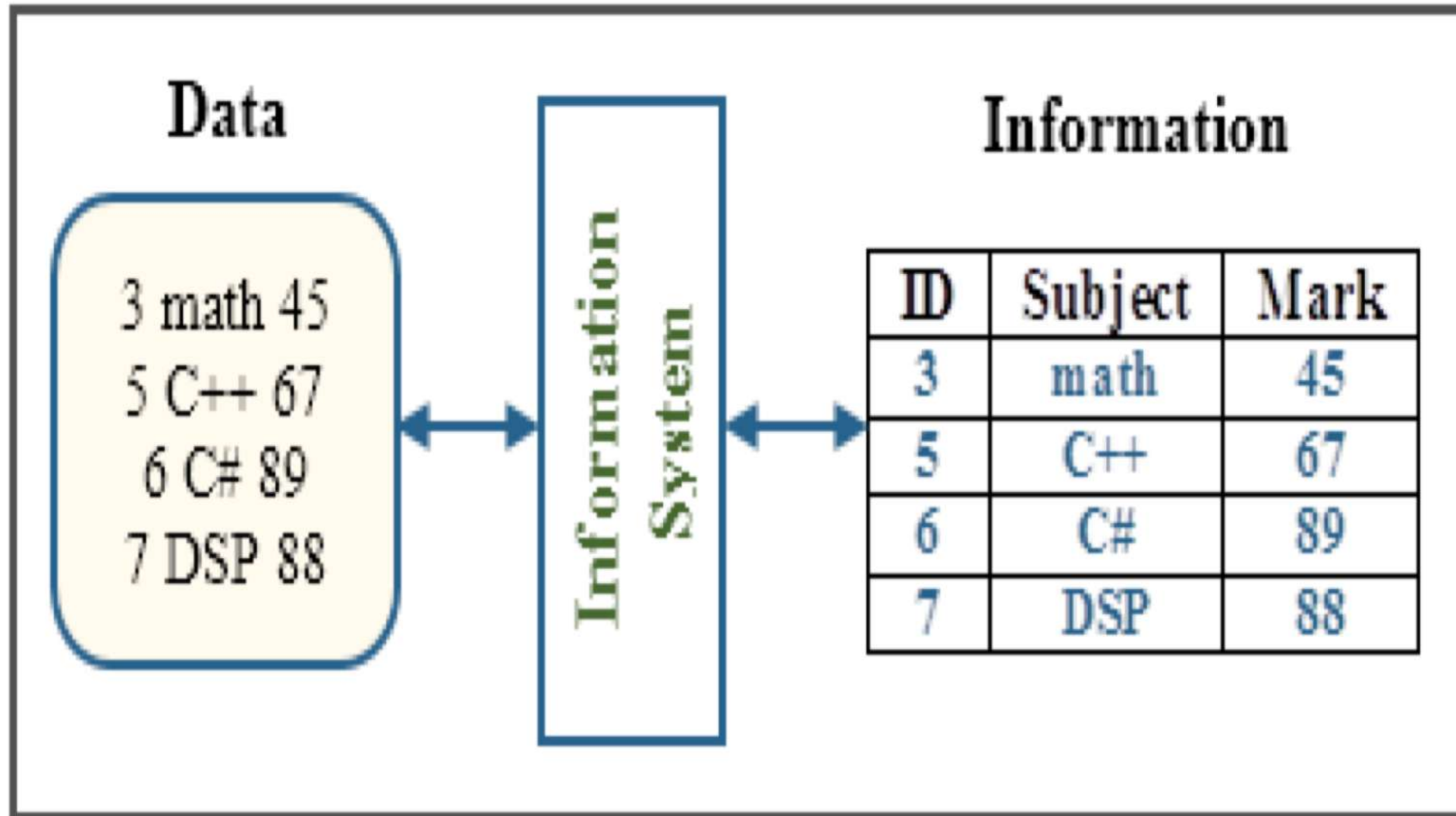
Information Vs Data

- ❖ Data can be described as **unprocessed facts and figures**.
 - Plain collected data as **raw facts** cannot help in decision-making.
- ❖ Information is **interpreted data**; created from **organized, structured, and processed data** in a particular context.
 - **Information** is a **data that has been processed** into a form that is **meaningful** to recipient and is of real or perceived value in the current or the prospective action or decision of recipient.



Processing involves selecting, organizing, and manipulating data

Information Vs Data typical Example



INFORMATION CHARACTERISTICS

❖ What is Information?

- **Information** comprises the meanings and interpretations that people place upon facts.
- The value of information springs from the ways it is interpreted and applied to make products, to provide services, etc.

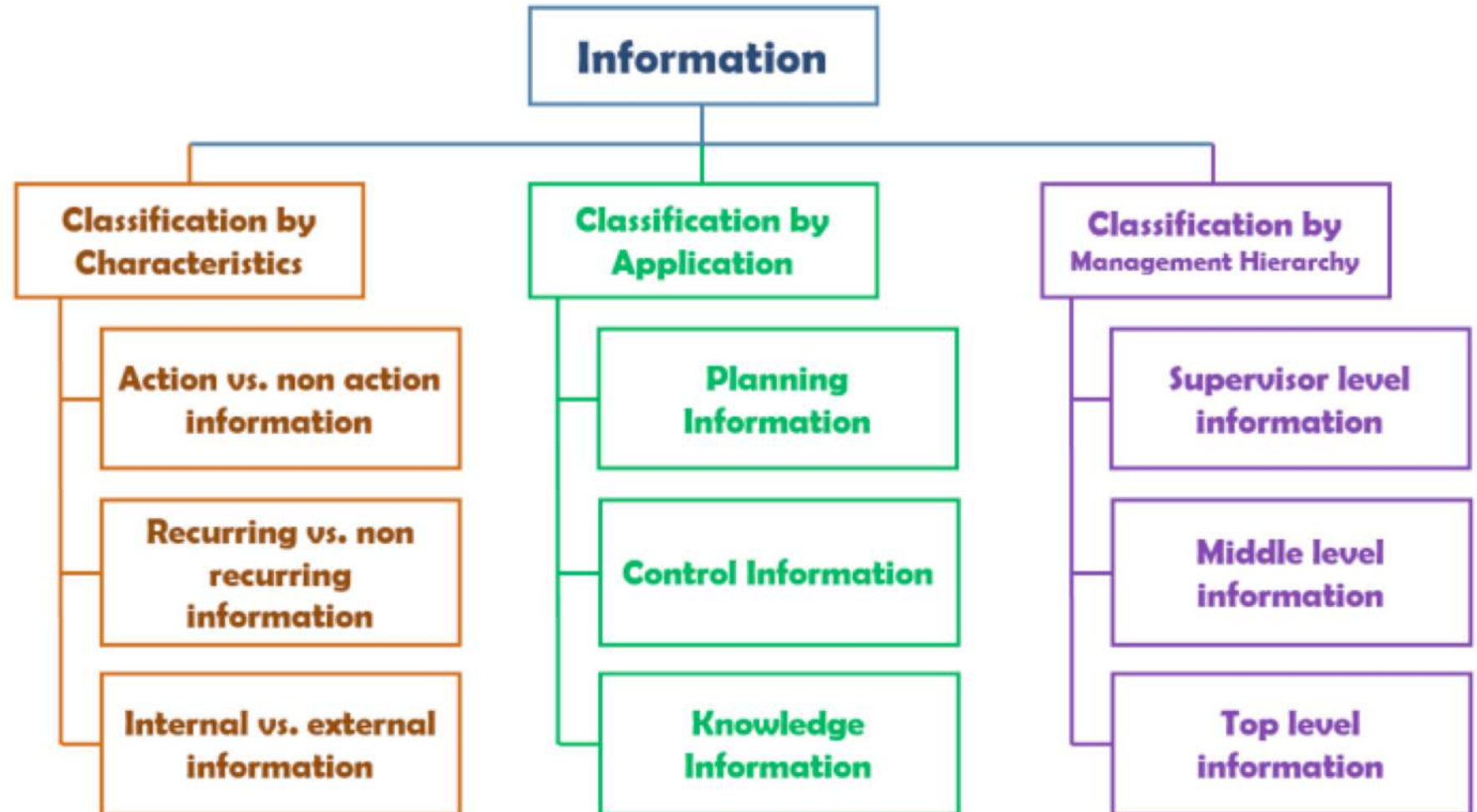
❖ Information has three characteristics:

- 1. It can be recorded and retrieved;
- 2. It has value & meaning.
- 3. It can exist in many forms, e.g., written, printed, electronically stored, physically transmitted or transmitted in electronic form.

❖ A database is an organized collection of structured information, or data, typically stored electronically in a computer system.

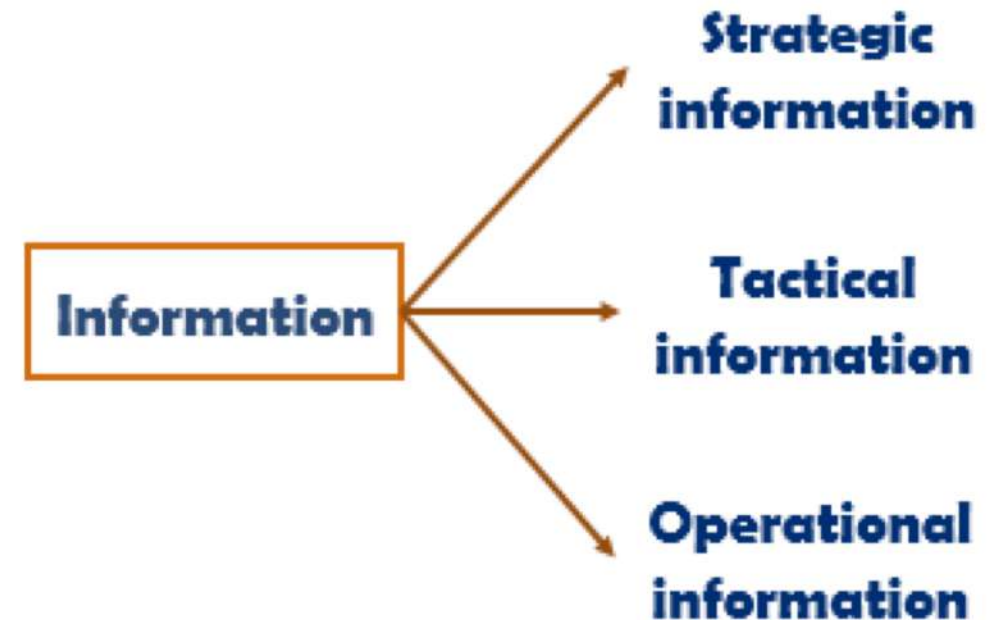
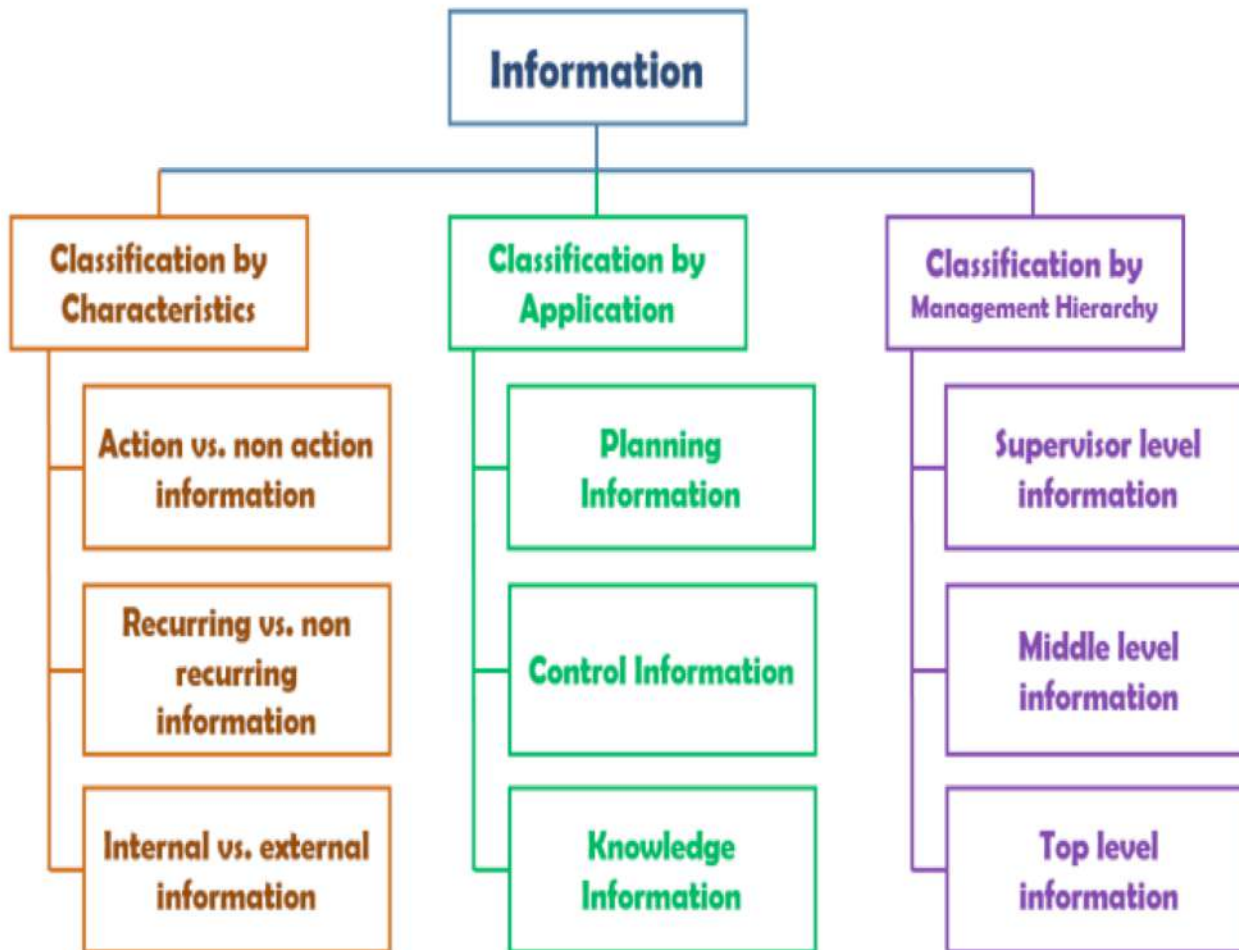
Classification Of Information

- ❖ Information can be classified in a number of ways, you will learn two of the most important ways to classify information.



Classification based on the decision making

- ❖ Based on Anthony's, from the view of management classification of information that is gathered for the purpose of decision making can be classified as shown in figure 4 below:

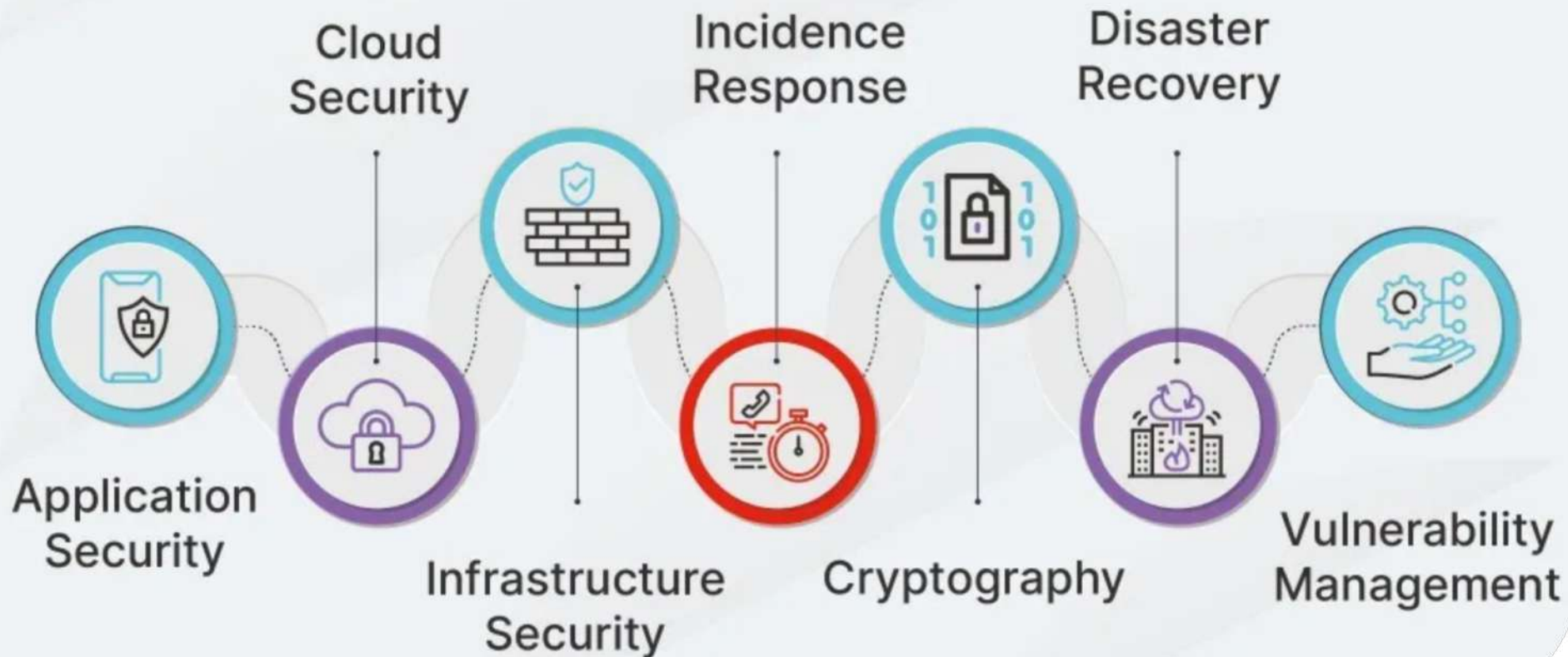


Information Security?

- ❖ **Information Security** is the **practice** of preventing **unauthorized access**, **use**, **disclosure**, **disruption**, **modification**, **inspection**, **recording** or **destruction** of information.
- ❖ This is to mean, even **though information is important to all of us**, **not all information should be accessed by anyone**.
 - **The access** should be controlled.
 - Every organization **uses information**, most are dependent on it.
- ❖ Information is an asset, which can be: **Business records**, **Client and contact databases**, **Personnel information**, **Financial records and transactions**, **E-commerce transaction details**.

TYPES OF INFORMATION SECURITY

Types of Information Security



Basic security components

- Confidentiality
- Integrity
- Availability



1. Confidentiality

- ❖ Means information is **not disclosed to** unauthorized individuals, entities and process. For example:
 - If we say **I have a password for my Gmail account** but someone saw while I was doing a login into Gmail account. In that case **my password has been compromised and Confidentiality has been breached**.
 - Someone watching you type in **your pin code at an ATM**, is compromising the confidentiality of that pin code.
 - If you leave your **PC while you are still logged on**, and someone **starts browsing through your files in your absence**, this would be considered as an **attack on confidentiality** of your information.

2. Integrity

- ❖ Used to indicate whether information has **been changed** or whether that information **has errors** in it.
 - So, someone who **changes the data in a file without proper authorization** will have compromised the **integrity of that data**.
 - The **most common attack on integrity**, however, **simply consists of input errors** when that data/information is first put on a system or is updated.
- ❖ **Examples:**
 - If an employee **leaves an organization**, then in that **case data for that employee** in all departments like **accounts**, **should be updated** to reflect status to **JOB LEFT** so that data is complete and accurate and only authorized person should be allowed to edit that employee's data.
 - Attacks on the integrity of information can have **an objective fraud** (e.g., falsifying the records in a payroll system) or **a political motive** (e.g., modifying the contents of a website).

3. Availability

- ❖ Its used to indicate whether a user has **access to any information**, or the system that contains/processes that information, when the user needs this access.
- ❖ **An attack on availability** can involve **deleting information or crashing a system**, but it may **also prevent access to the system** by **cutting or overloading** the communication channel to that system.
 - The most common **attacks on availability** are, however, **simple things like hardware failure or power failure**.

Information Security Controls

❖ Types of information security controls include:

- Authentication
- Encryption
- Integrity controls
- Backups
- Access control
- Auditing

1. Authentication

- ❖ **Authentication** is the process or action of **verifying the identity** of a user or process.
- ❖ is the **process of Ensuring** that both ends of the connection are in fact **who they say they are**.

Authentication: Three Types

❖ Single Factor Authentication

- Password
- Easy to remember
- Easy to crack
- People are predictable...passwords are usually a pets name, a birth date, etc.

❖ Two Factor Authentication

- Password + **token** (security device for users to keep in possession)
- Safer and more complex than single factor

Cont...

❖ **Three factor Authentication**

- Password + token + **biometric authentication** (fingerprint, retinal scan)
- Safer and more complex than single or double factor types;
- used for high security purposes (ex. Government documents)

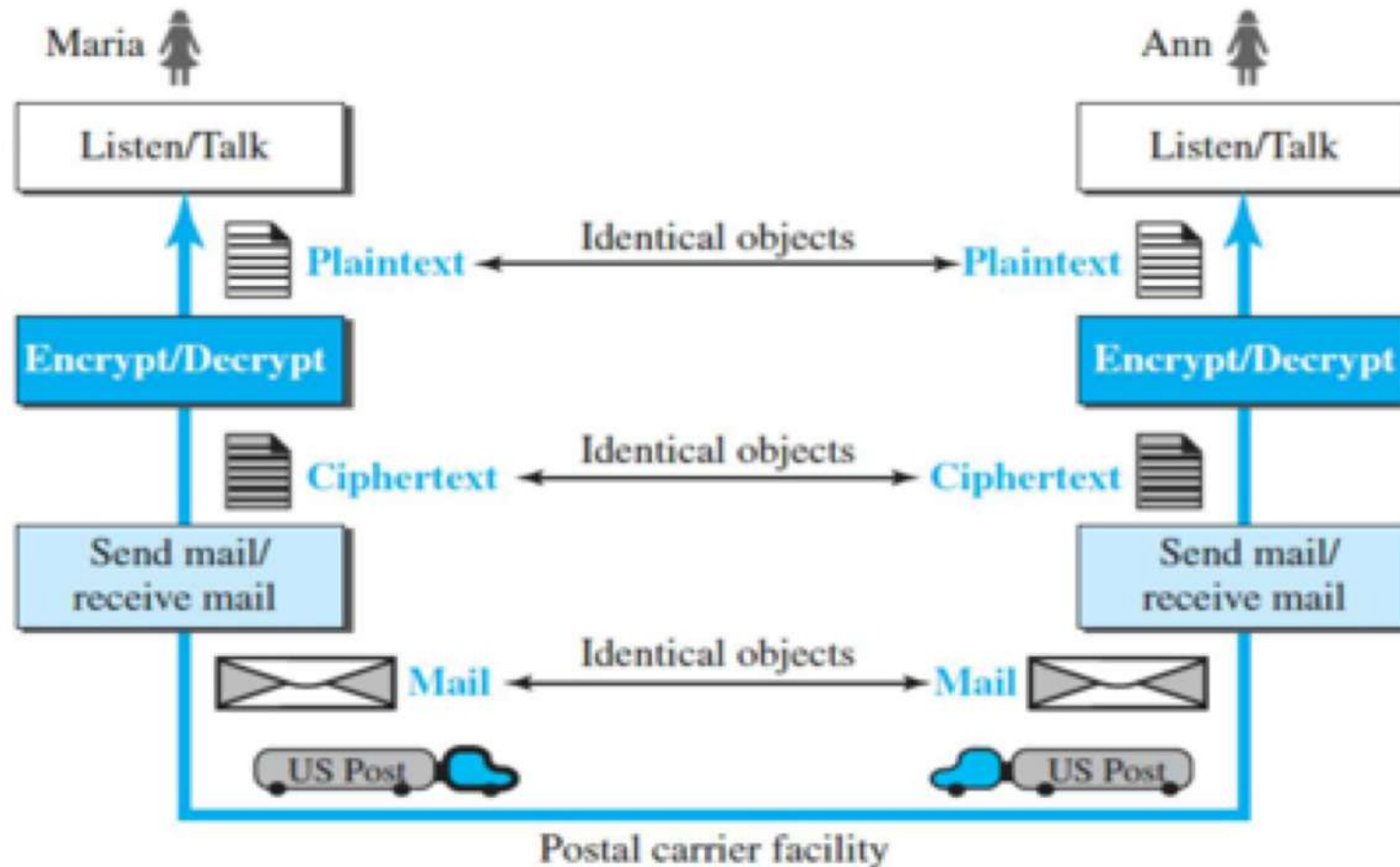
❖ **A token is a security device for authorized users to keep in possession.**

❖ Some examples include:

- **Security ID Card, Challenge/response method, and USB token**

2. Encryption and Security (Cryptography)

- ❖ **Encryption** is a mechanism for hiding information by turning readable text (**PLAIN TEXT**) into a stream of digital signatures (**CIPHER TEXT**) in such a way that someone with the proper key can make it readable again.



3. Integrity controls

- ❖ Are designed to **manage the integrity of data**, which is a fundamental component of information security.
 - ❖ In its broadest use, “**data integrity**” refers to the **accuracy and consistency of data stored in a database, data warehouse, data mart, or other construct**.
 - ❖ This can be used by adopting:
 1. **scans,**
 2. **bar code,**
 3. **digital signatures, etc.**
-

4. Access Controls

❖ is a **security technique** that regulates **who or what can view or use resources** in a computing environment.

❖ Authentication

- **identity confirmation**

❖ Authorization

- permission often role-based
- **Firewalls**

❖ Accountability

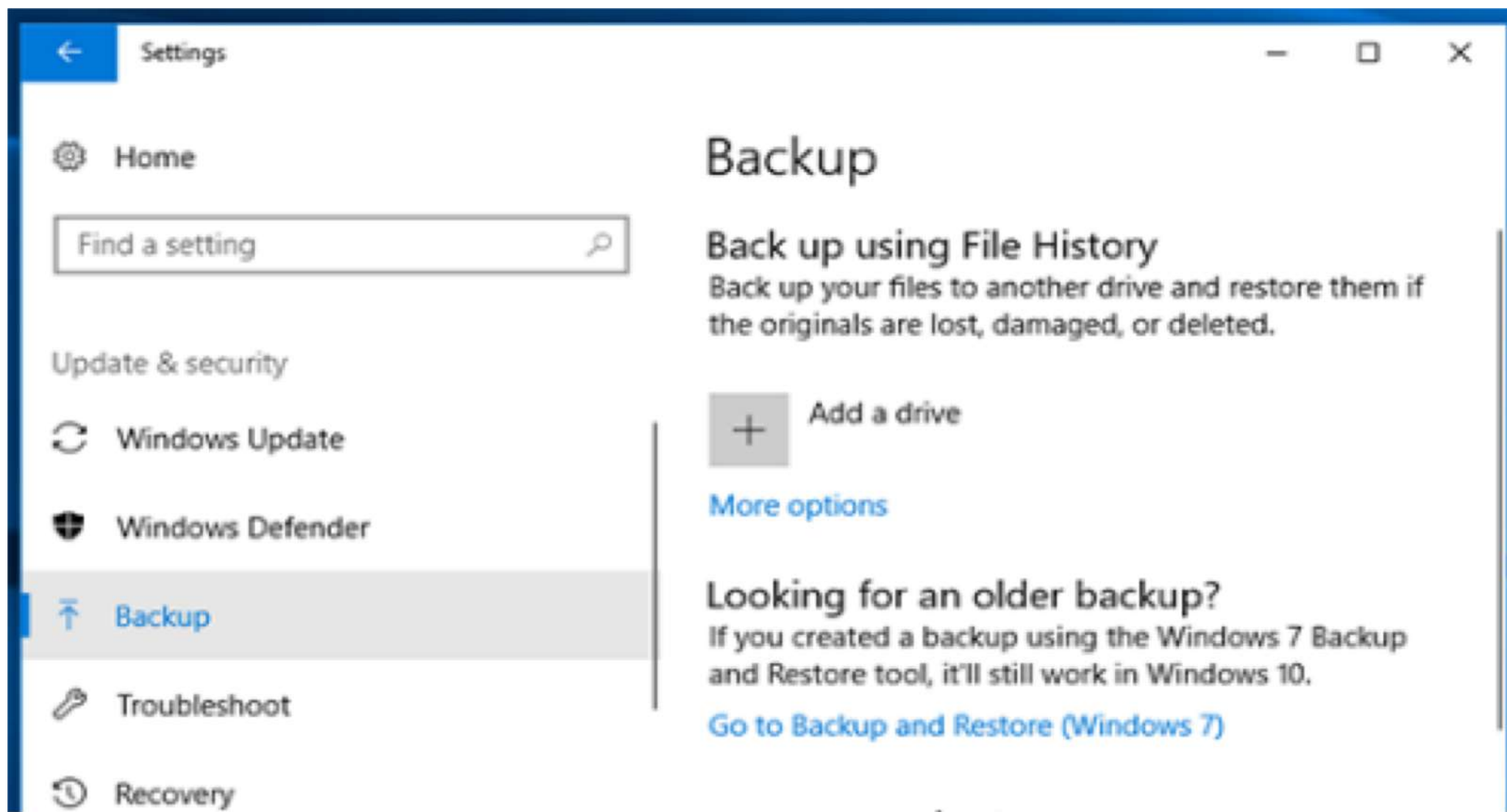
- logging



5. Backup

- ❖ **Data backup** is a **copy of computer data taken and stored** elsewhere so that it may be used to **restore the original** after a data loss event.
 - ❖ You can back up either to:
 - 1. **Backing up to an external device**, such as **tape, HDD**, and then it can be transferred and stored on **another device** in case of data loss.
 - 2. **Backing up to the cloud** and then opening it on another computer by logging into the **cloud account**.
-

Back Up And Restore in Windows



Reason for database security

- ❖ Guard firm's reputation
- ❖ Avoid litigation(controversies)
- ❖ Retain competitive standing
- ❖ Maintain trust
 - Customers
 - Merchants
 - Business partners/vendors

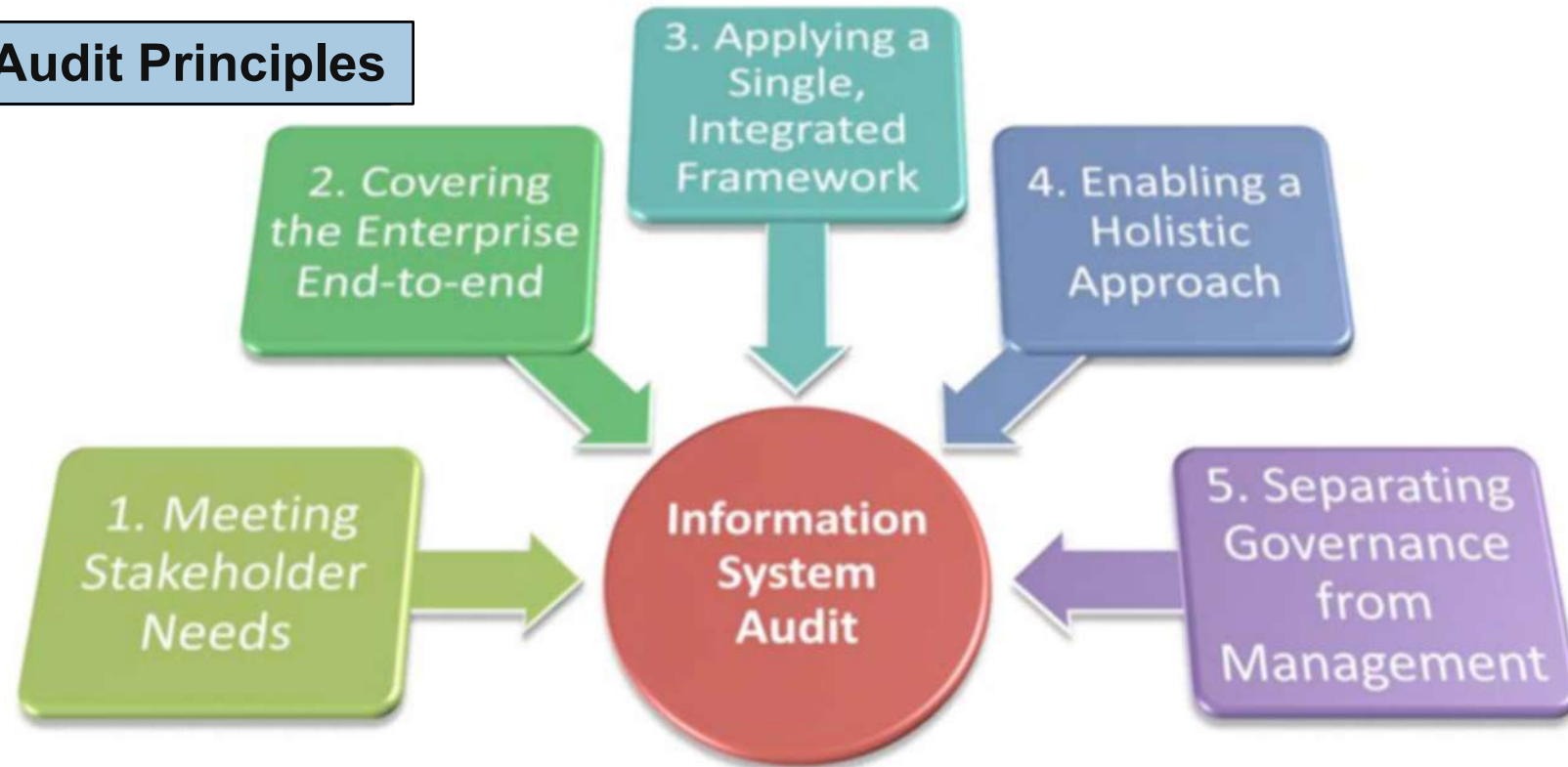
DISCOVER
NETWORK



6. Information Auditing (IA)

- ❖ The role of the information audit is to provide a method for identifying, evaluating, and managing information resources in order to fully exploit the strategic potential of information.
- ❖ Information is representative of a resource which requires effective management and this led to the development of interest in the use of an IA.

Information System Audit Principles

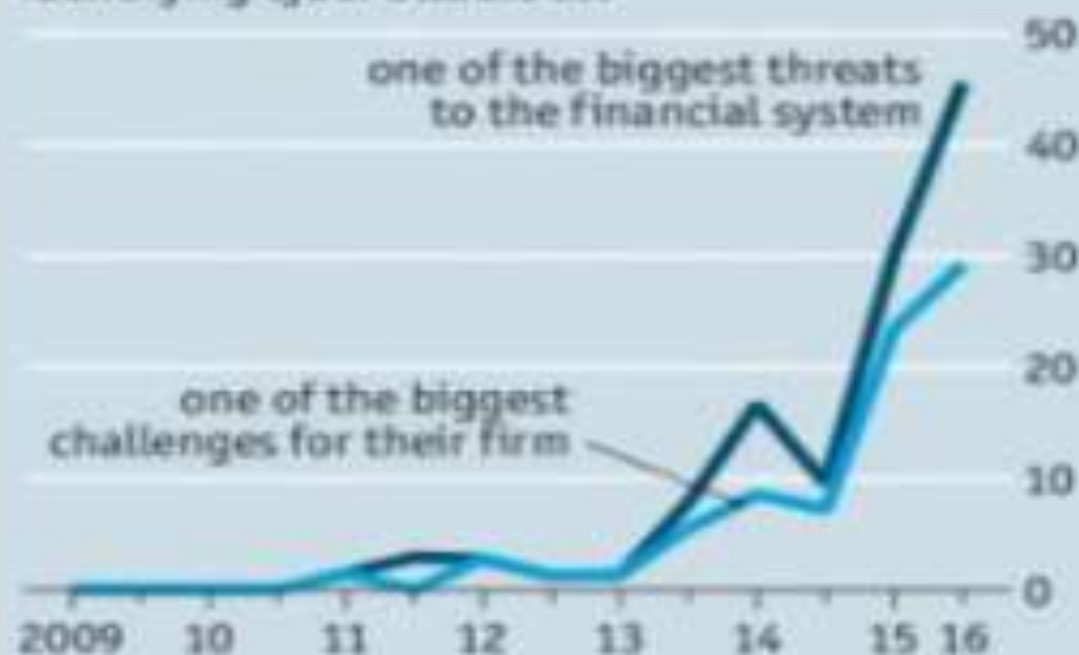


WHY INFORMATION SECURITY?

Cyber attacks on Financial Industry

Three-alarm firewall

% of British finance executives identifying cyber attacks as:



Source: Bank of England Systemic Risk Survey

Technology

Cyber attacks against financial services cost consumers £8bn in 2016, research reveals

The Guardian

The time a major financial institution was hacked in under 15 minutes

GCHQ asked to step up action against cyber-attack threat to financial services

Treasury select committee says existing vulnerabilities and accountability need to be addressed in the wake of Tesco Bank hacking

QUIZ - DISCUSSION

1. If you receive an e-mail from someone you don't know, what should you do to ensure the protection of your data?
2. Discuss the security for paper vs Electronic Records?
Discuss the pro's and con's of paper vs electronic record keeping in terms of security.
3. Information is considered as an asset in this digital era.
Agree or disagree. Discuss

QUIZ - DISCUSSION

1. Why information systems are vulnerable NOWADAYS as compared to back then in 20th century? Discuss in details.
2. The mouse on your computer screen starts to move around on its own and click on things on your desktop. What do you do?
3. If you receive an e-mail from someone you don't know, what should you do to ensure the protection of your data?
4. Discuss the security for paper vs Electronic Records? Discuss the pro's and con's of paper vs electronic record keeping in terms of security.
5. Information is considered as an asset in this digital era. Agree or disagree. Discuss

What is Information security management?

- ❖ **Information security management** is the **process** by which **the value of each of an organization's information assets** is **assessed** and, if appropriate, **protected** on an ongoing basis.
 - We can deduce that **no single solution** can address all possible **security concerns**.
 - The **only strategy** is to engineer **a fit-for-purpose solution** that achieves a **suitable balance between risks and protection** against them.
- ❖ How do we move forward?
 - **Implementing information security** in an organization can protect the technology and information assets it uses by **preventing, detecting and responding** to internal and external threats.

Security risks to database systems

- ❖ **A database** is an **organized collection** of **structured information, or data**, typically **stored electronically** in a computer system.
- ❖ **Unauthorized** or **unintended activity** or **misuse by authorized database users**,
 - database administrators,
 - or network/systems managers,
 - or by unauthorized users or hackers (e.g. inappropriate access to sensitive data, metadata or functions within databases,
 - or inappropriate changes to the database programs, structures or security configurations)

Cont.

- ❖ **Malware infections** -software that is specifically designed to **disrupt, damage, or gain unauthorized access** to a computer system.
- ❖ causing incidents such as:
 - **unauthorized access,**
 - **leakage or disclosure** of personal or proprietary data,
 - **deletion of or damage** to the data or programs,
 - **interruption or denial of authorized access** to the database,
 - attacks on other systems and the unanticipated failure of database services;

Cont.

- ❖ **Overloads**, performance constraints and capacity issues resulting in the inability of authorized users to use databases as intended;
- ❖ **Physical damage to database servers** caused by:
 - computer room fires or floods,
 - overheating,
 - lightning,
 - accidental liquid spills,
 - static discharge,
 - electronic breakdowns/equipment failures and
 - obsolescence;

Cont..

- ❖ **Design flaws and programming bugs in databases** and the associated programs and systems. This can create **various security vulnerabilities** such as:
 - unauthorized privilege escalation,
 - data loss/corruption,
 - performance degradation etc.
 - ❖ **Data corruption and/or loss** caused by the **entry of invalid data or commands, mistakes** in database or system administration processes, **sabotage/criminal damage** etc.
-

Thank you for Your Attention

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